



Wednesday Feb-26-2025

Flowserve Facility:

Flowserve Nordstrom / Valbart Valves
1511 Jefferson St.
Sulphur Springs, TX 75482

cmtrs@flowserve.com

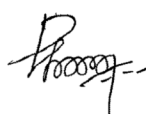
To:

CERTIFICATE OF CONFORMANCE

SERIAL #	HEAT #	PO #	Description	Class	End Connection
61-677522	2629Z	746235	4" 2345	900	FLG
61-665690	0462Z	738357	4" 2345 RTJ	900	FLG

We hereby certify that all the subject valves/parts/sealant have been manufactured in conformance with the applicable requirements of Flowserve Sulphur Springs Operation (FSSO) specifications and that the valve/parts/sealant materials for component parts meet the applicable requirements of FSSO drawings and ASTM specifications, ASME B16.34 (If Applicable), API Q1 10th edition, and API 6D 25th edition.(if applicable to the API monogram). Flowserve Dynamic Balanced Valves are provided with a Pressure balance feature unless specifically identified in the Valve description. All valves/parts/sealant meet all the requirements of the customer purchase order.

Documentation Technician
Justin McKee

	FLOW LINK SYSTEMS PRIVATE LIMITED FOUNDRY UNIT - II 65 / 2-A,B & C, PUDUPALAYAM B.P.O AVINASHI TK, COIMBATORE - 641 654 INDIA		INSPECTION CERTIFICATE (EN 10204:2004-3.1)		TC NO : X-3054 DATE : 12-Nov-2022									
	CUSTOMER FLOWSERVE INDIA CONTROLS PVT LTD, B8, MMDA Industrial Area, Maraimalai Nagar-603209 Kancheepuram, INDIA		SPECIFICATION REFERENCE 1. ASTM A216-21 Gr.WCC 2. Customer Spec. RMC 90012 Rev. 14		HEAT NO. 2629Z Date of Pouring 24-Sep-2022 FOUNDRY MARK FLS									
PO NUMBER & DATE 10107157 Dt. 06.09.2022			ORDER ACCEPTANCE NO 22-1986											
CHEMICAL COMPOSITION % wt														
MELTING PRACTICE : Steel melted using Electric Induction furnace, and fully killed using Aluminium.														
SPECIFICATION	C	Si	S	P	Mn	Ni	Cr	Mo	V	Cu	Al	--	*RE	**CE
Minimum	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Maximum	0.23	0.60	0.020	0.025	1.20	0.50	0.50	0.20	0.030	0.30	0.08	--	1.00	0.43
Achieved	0.183	0.377	0.006	0.017	0.958	0.057	0.095	0.011	0.006	0.016	0.040	--	0.19	0.37
Heat Treatment : NORMALIZING: Temperature raised to 920°C, Soaked for 4 hours and then atmospheric air-cooled														
Visual Inspection - Castings conforms to ANSI/MSS-SP-55-2011 - Type II to XII (a)														
MECHANICAL PROPERTIES														
TEST TYPE	Tensile as per ASTM A370-21 / ISO 6892-1-2020						Impact (CVN) ISO 148-2016/ASTM A370-21		BEND TEST		HARDNESS			
SPECIFICATION	YS Rp 0.2% (KSI)	YS Rp 1.0% (KSI)	UTS (KSI)	% Elongation		% RA	Test Temperature (at °C) J		ANGLE	D	(HBW / HRBW)			
				4d	5d									
							Values	Avg.						
Minimum	40	--	70	22	--	35	--	--	--	--	--			
Maximum	--	--	95	--	--	--	--	--	--	--	--			
Achieved	45.6	--	75.6	33.6	--	64.8	--	--	--	--	150			
DETAIL OF CASTINGS														
S.No.	PO S.No.	DESCRIPTION	CAST DRAWING NO / REV	PART NO		POURED QTY	DESP. QTY							
1	02	4" 900 FE BODY	D-487876 / 20	N2757930-60		3	1							
MTC - REVIEWED & ACCEPTED  10139322 FICPL QC														
REMARKS: 1. Stated Chemical composition is from heat analysis; and Mechanical properties of test coupon from Cast keel block. 2. *RE = Ni + Cr + Mo + V + Cu $** CE = C + \frac{Mn}{6} + \frac{Cr+Mo+V}{5} + \frac{Ni+Cu}{15}$ 3 4. The above stated KSI values have been calculated based on original tested N/mm² values.														
CERTIFIED THAT 1. THE ABOVE GIVEN DETAILS ARE CORRECT, FULFILLS THE SPECIFICATION / STANDARDS AND PO. 2. MATERIAL WAS MANUFACTURED, SAMPLED, TESTED, INSPECTED IN ACCORDANCE WITH THE ABOVE SPECIFICATIONS AND CERTIFIED AS PER EN 10204 - 3.1. 3. THE MATERIAL IS FREE OF CONTAMINATION FROM RADIO ACTIVE ELEMENT OR RADIATION.				 M.S.DHANABALAN ASSISTANT MANAGER - QUALITY MANUFACTURER'S AUTHORIZED INSPECTOR										
PREPARED BY:  S.KARTHICK		CHECKED BY: A.LOGANATHAN LAB CHEMIST 												